Ordinary Kriging Surrogates in Aerodynamics

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Introduction

Surrogate models become increasingly popular in performing various optimization or uncertainty quantification (UQ) analyses. The principle of a surrogate model relies on an efficient approximation method which estimates a scalar or vector-valued input/output functional using a data set constituted by observations of this functional, often computationally expensive [1]. For example, the computational resources necessary to evaluate stochastic integrals are known to grow exponentially with the number of dimensions due to the so-called curse of dimensionality. This increase becomes even more critical when the integrands need to be evaluated by intensive computations. Surrogate models are a non-intrusive alternative for such

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computations which can outcome the limitation emphasized before by providing cheap representations of the integrands in the parameters space. These representations are typically computed using an interpolation or a regression procedure. However, when considering a moderate to large parameters space, efficient algorithms are mandatory to derive accurate representations of parameterized integrands.

The Kriging procedure [2, 3] is a good candidate because of its robustness, accuracy, and ability to provide a description of the error done by replacing a complex function by a Kriging surrogate. This approach is interpolating and assumes that the function to approximate is the realization of a second-order Gaussian process. The construction of this interpolation using a sampled data set is based on a covariance function, often chosen as Gaussian, of which inner parameters have to be tuned up in order to provide a reliable surrogate. Here, we apply this procedure to the simulation of airflow around a wing profile with the consideration of variable shape parameters of the profile. The present study is more particularly dedicated to a two-dimensional RAE 2822 airfoil simulated at transonic speed [4, 5]. It is a popular, well-documented test case in the literature and has received much attention over the past decades; see [6, 7] and references therein for an application to the development of algorithms for shape optimization. Our objective is to obtain an accurate surrogate for evaluating the aerodynamic performance of the airfoil when its shape is altered, considering a dense data set with a moderately high dimension of the parameters space (four in the present investigation). This accuracy is required for a subsequent robust optimization of the airfoil. We note at this stage that the Kriging procedure becomes computationally intractable for high-dimensional parameters space and large data sets, and alternative techniques need be advocated in these situations as emphasized in [8]. They include, for example, sparse grid-based polynomial projection or regression techniques using either structured or unstructured nodes, as detailed in the previous chapters "General Introduction to Polynomial Chaos and Collocation Methods" and "Generalized Polynomial Chaos for Non-intrusive Uncertainty Quantification in Computational Fluid Dynamics" and references therein.

The rest of this chapter is organized as follow. The next section "Test Problem" introduces the parameterized test problem considered in this work and the numerical tools used to construct the data set. Details about the Kriging method are presented in section "Kriging Surrogate Model." The inputs and results of a robust UQ study based on a Kriging surrogate are described in section "Robust Optimization Based on the Surrogate Model." Finally, some conclusions are drawn in section "Conclusions."

Test Problem

Actual cruise flight conditions are mostly transonic, meaning that the flow can become locally supersonic due to the geometry of the wing profiles. Its acceleration on the upper surface of the profiles induces a depression which is responsible for the lift force. If this depression is too sharp, a discontinuity, or shock wave, is created to balance the pressure gradient at the trailing edge between the upper and lower surfaces. The position and strength of the shock wave are responsible for a part of the drag force, and hence, these features are of major importance in optimization, for instance to minimize the drag force at constant lift force. Modifications of the shape of the profiles can alleviate such issues by smoothing out the discontinuity, hence increasing the lift force while decreasing the drag force. We thus consider a test problem where the objective is to develop a numerical strategy to quantify the influence of variable geometrical parameters of an airfoil on its aerodynamic performances such as the lift and/or drag force.

More particularly, the present test case is a transonic turbulent flow around a RAE 2822 airfoil. The flow is modeled by the steady-state Reynolds-averaged Navier–Stokes (RANS) equations together with a Spalart–Allmaras turbulence model closure [9], which are now routinely used for the quantification of design issues involving compressible aerodynamics. The baseline conditions of the flow are described in [4] for the test case #6 together with the wall interference correction formulas derived in [10, pp. 386–387] and their slight modifications suggested in [11, p. 130]. The operational parameters considered here are thus $M_\infty = 0.729$ for the free-stream Mach number, $\alpha_\infty = 2.31°$ for the angle of attack, and $Re = 6.50 \cdot 10^6$ for the Reynolds number based on the chord length c, fluid velocity, temperature, and molecular viscosity at infinity. They arise from the corrections $\Delta M_\infty = 0.004$ and $\Delta\alpha_\infty = -0.61°$ given in [11, p. 130] for the test case #6 outlined in [4], for which $M_\infty = 0.725$, $\alpha_\infty = 2.92°$, and $Re = 6.50 \cdot 10^6$.

Discretization and Numerical Parameters

In this section, we describe how the numerical simulations of the foregoing example are performed. The flow is computed using the cell-centered finite volume CFD software *elsA* [12]. The RANS equations are discretized using a $641c \times 129c$ mesh, shown in Fig. 1, where the boundary at infinity was left intentionally far at $450c$ from the airfoil. These values proved to be sufficient to avoid spurious reflections with the far-field boundary. The discretized numerical solution is obtained based on:

- A Roe flux using a second-order MUSCL scheme [13] (based on van Albada limiter [14]) for the convective term of the RANS system;
- First-order Roe fluxes for the advection term of the turbulent variable;
- Corrected second-order diffusive terms based on the corrected mean of the cell-centered gradients of the two adjacent cells;
- Source terms for the turbulent transport computed using the temperature gradients at the center of the cells.

The flow is attached with a weak shock wave on the suction side. The static pressure at the wall as well as the iso-Mach number levels are presented in Fig. 2. Given the large number of simulations to run, the parameters of the steady-state algorithm proved to be essential to insure a fast convergence. This was achieved using the following set of parameters:

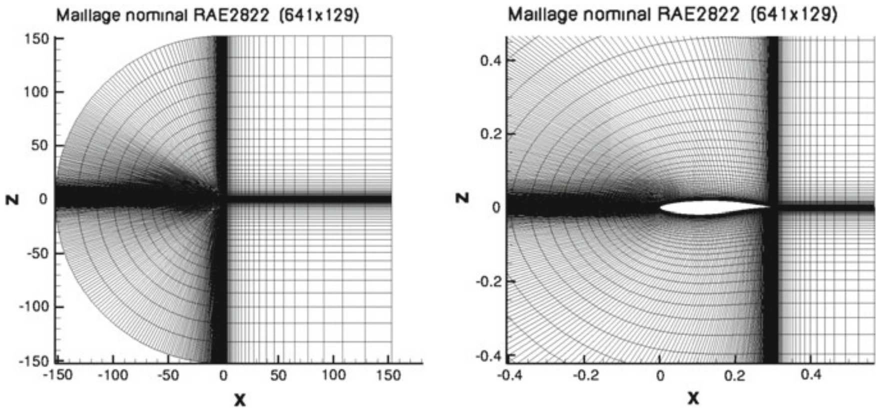

Fig. 1 Computational domain (left) and close view (right) of the mesh for the baseline RAE 2822 configuration

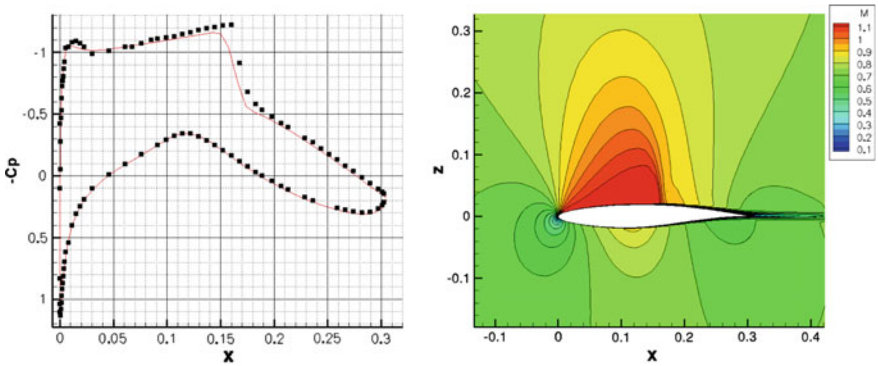

Fig. 2 Static pressure coefficient $-C_p$ at the wall (left) and iso-Mach number levels (right) for the baseline RAE 2822 configuration at $M_\infty = 0.729$, $\alpha_\infty = 2.31°$, $Re = 6.50 \cdot 10^6$ compared with the experimental data reported in [5]

- An implicit Lower-Upper Symmetric Successive Overrelaxation (LU-SSOR) scheme [15] using four relaxation cycles, increasing the CFL conditions after 100 iterations to CFL = 50;
- A multi-grid approach for the Navier–Stokes system over two grid levels with two iterations on the coarse grid;
- A single fine level iteration for the turbulent equation alternating with a multi-grid iteration for the RANS system.

The corresponding decrease of the explicit residual for the mass flow is shown in Fig. 3 (left). Once the numerical parameters have been fixed, the number of iterations is determined from the evolution of the global forces, through a 30,000 iterations calculation (decreasing discrete residuals of all equations being checked at every iteration, the final reduction of the mass conservation residual, shown in Fig. 3, is

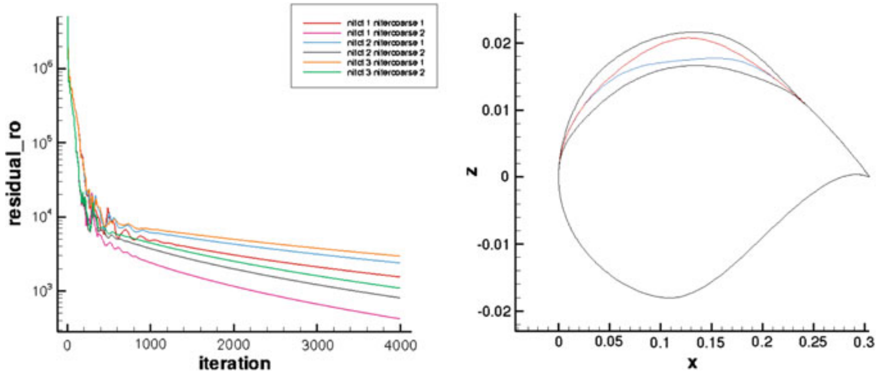

Fig. 3 Convergence of the multi-grid simulations (one level of coarse grid) for the baseline mesh (left) and maximum and minimum displacement (black lines) of the shape of the RAE 2822 profile using the four parameters of the present study (left). Maximum (red) and minimum (blue) displacement of the third bump alone

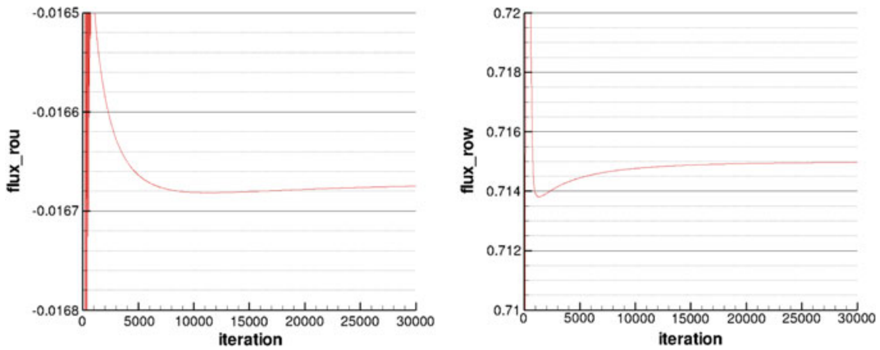

Fig. 4 Total forces along the x horizontal axis (left) and z vertical axis (right) as a function of the number of iterations

$2.5 \cdot 10^4$). After 6000 multi-grid cycles, the force values appeared to remain within the range $[-2/10{,}000, +2/10{,}000]$ which is acceptable, given the convergence error; see Fig. 4. Hence, this number of iterations has been retained for all subsequent calculations.

Shape Parameters

The shape of the baseline RAE 2822 profile is subsequently altered using four B-spline functions, located on the suction side of the airfoil. These alterations are responsible for triggering the pressure gradient induced by the geometry of the profile. The pressure gradient is significantly altered even for small variations of the airfoil shape, which leads to nonlinear variations of the lift and drag forces. The shape parameters used to alter the baseline configuration are:

- Four control points located at 5, 20, 40, and 60% of the chord c;
- The deformation is described by B-spline functions denoted by $S_n(s)$ such that:

$$S_n(s) = \sum_{j=0}^{m-n-1} \alpha_j b_n^{(j)}(s)\,, \tag{1}$$

  where the α_j's are the values of the deformed shape at the control points and s is the curvilinear abscissa of the airfoil. The $m - n$ basis functions $b_n^{(j)}$ of degree n are defined by the recurrence relation:

$$b_0^{(j)}(s) := \begin{cases} 1 & \text{if } s_j \le s < s_{j+1}\,, \\ 0 & \text{otherwise,} \end{cases}$$

  and:

$$b_n^{(j)}(s) := \frac{s - s_j}{s_{j+n} - s_j} b_{n-1}^{(j)}(s) + \frac{s_{j+n+1} - s}{s_{j+n+1} - s_{j+1}} b_{n-1}^{(j+1)}(s)\,.$$

  Here, $m = 4$ and $n = 2$ have been chosen;
- The amplitude ξ_k, $k = 1, 2, 3$, or 4, of each B-spline varies in the range $\pm 0.0025c$ in the direction of the outward normal vector to the nominal (unaltered) profile;
- The leading edge and the trailing edge are considered as fixed.

The shape of the airfoil thus evolves continuously as a function of the shape parameter vector $\boldsymbol{\xi} := (\xi_1, \xi_2, \xi_3, \xi_4)$ which varies in the design domain $D = [-0.0025c, 0.0025c]^4$. The support of each spline independently of the others is depicted in Figs. 5 and 6. The effect of the maximum and minimum displacements is shown in Fig. 3 (right).

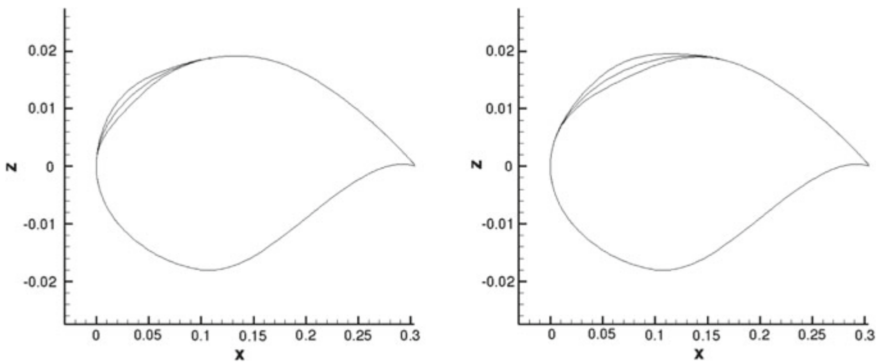

Fig. 5 Baseline configuration and deformation of each spline with an amplitude in the range $[-0.0025c, 0.0025c]$ at $0.05c$ (left) and $0.2c$ (right)

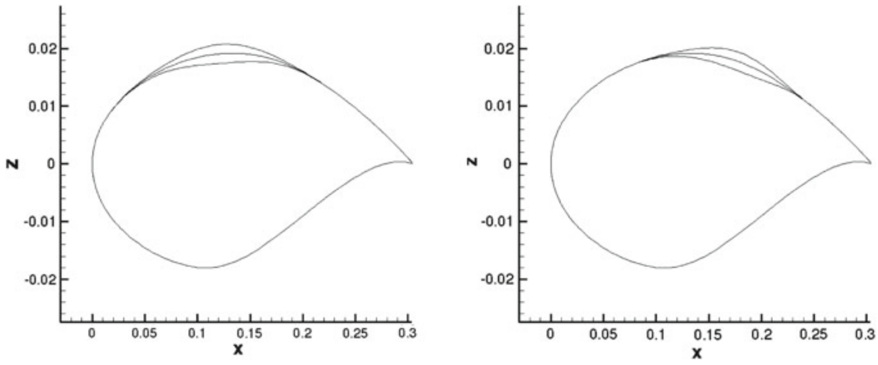

Fig. 6 Baseline configuration and deformation of each spline with an amplitude in the range $[-0.0025c, 0.0025c]$ at $0.4c$ (left) and $0.6c$ (right)

Quantity of Interest

The quantity of interest is the lift-to-drag ratio Y where the lift coefficient is denoted by C_L and the drag coefficient by C_D, such that:

$$Y(\boldsymbol{\xi}) = \frac{C_L(\boldsymbol{\xi})}{C_D(\boldsymbol{\xi})}\,. \tag{2}$$

Note that in the present work the computation of the drag coefficient has been performed using the far-field approach detailed in [16]. Our aim is thus to provide an optimal approximation (in a sense clarified in the subsequent section) for the function $\boldsymbol{\xi} \mapsto Y(\boldsymbol{\xi})$. This is achieved on the basis of $Q > 0$ repeated runs of the CFD software for the altered configurations at selected parameter values $\boldsymbol{\xi}^l = (\xi_1^l, \xi_2^l, \xi_3^l, \xi_4^l)$, $1 \leq l \leq Q$, in the normalized parameter domain $\hat{D} = [0, 1]^4$. The data set containing all the simulations is denoted by $S_Q = \{Y(\boldsymbol{\xi}^l);\ 1 \leq l \leq Q\}$ in the following, with $Q = \#S_Q$.

Kriging Surrogate Model

This section outlines the numerical procedure used to compute an optimal interpolant, or surrogate model, of the quantity of interest Y by the Kriging method [1–3, 17, 18]. It also describes the strategy retained to obtain optimized inner parameters for the Kriging interpolant. For that purpose, we basically follow the presentation in [8]. As already stated above, our aim is to use this surrogate to characterize the influence of alterations of the baseline profile of the RAE 2822 airfoil described in section "Shape Parameters" on its aerodynamic performance, as expressed here by the ratio between the lift and drag coefficients (2).

The Kriging Method

The Kriging method is based on the assumption that the quantity of interest $Y(\boldsymbol{\xi})$ is a random variable which can be decomposed into [1]:

$$Y(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi}) + \varepsilon(\boldsymbol{\xi}), \tag{3}$$

where $\mu(\boldsymbol{\xi})$ and $\varepsilon(\boldsymbol{\xi})$ are sought as a deterministic contribution and a random fluctuation, respectively. The mean μ is typically sought as a constant or a low-order polynomial—hence the following terminology [3] of: (i) simple Kriging if μ is constant and known a priori; (ii) ordinary Kriging if μ is constant but unknown; or (iii) universal Kriging if μ is an unknown polynomial of known order (a polynomial chaos expansion, for example, as in [19]). The random process ε indexed by the parameters $\boldsymbol{\xi}$ is assumed to be a second-order, centered, mean-square stationary process, or stationary covariance process in the terminology of [18], such that its mean $\mu_\varepsilon = \mathbb{E}\{\varepsilon(\boldsymbol{\xi})\}$ and variance $\sigma_\varepsilon^2 = \mathbb{E}\{\varepsilon(\boldsymbol{\xi})^2\}$ are, respectively, null and constant. In these definitions, $\mathbb{E}\{X\}$ stands for the average of the random variable X. Thus, its covariance function depends only on the difference $\boldsymbol{\xi} - \boldsymbol{\xi}'$, and what is more the lag distance $\|\boldsymbol{\xi} - \boldsymbol{\xi}'\|$, that is to say $\mathrm{Cov}\{\varepsilon(\boldsymbol{\xi}), \varepsilon(\boldsymbol{\xi}')\} = \mathbb{E}\{\varepsilon(\boldsymbol{\xi})\varepsilon(\boldsymbol{\xi}')\} = \phi(\boldsymbol{\xi}, \boldsymbol{\xi}') = \varphi(\|\boldsymbol{\xi} - \boldsymbol{\xi}'\|)$. Here, the covariance of two random variables X and Y is defined by $\mathrm{Cov}\{X, Y\} := \mathbb{E}\{(X - \mathbb{E}\{X\})(Y - \mathbb{E}\{Y\})\}$ and the variance of a random variable X by $\mathrm{Var}\{X\} := \mathrm{Cov}\{X, X\}$. In this framework, the approximation (or linear predictor) of $Y(\boldsymbol{\xi}^*)$ at the unobserved coordinates $\boldsymbol{\xi}^\star$, denoted by $\mathscr{I}_Q Y(\boldsymbol{\xi}^\star)$, is a linear combination of the samples $Y(\boldsymbol{\xi}^l) \in S_Q$ such that:

$$\mathscr{I}_Q Y(\boldsymbol{\xi}^\star) = \sum_{l=1}^{Q} \lambda^l(\boldsymbol{\xi}^\star) Y(\boldsymbol{\xi}^l), \tag{4}$$

where $\lambda^l(\boldsymbol{\xi}^\star)$ are weights. The aim of the Kriging method is then to derive the best linear unbiased predictor in the sense of a mean-square error. It can be sought as the solution of a constrained minimization problem where the objective function is the variance $\sigma^2(\boldsymbol{\xi}^\star) = \mathrm{Var}\{\mathscr{I}_Q Y(\boldsymbol{\xi}^*) - Y(\boldsymbol{\xi}^\star)\}$ with the unbiased constraint $\mathbb{E}\{\mathscr{I}_Q Y(\boldsymbol{\xi}^\star)\} = \mathbb{E}\{Y(\boldsymbol{\xi}^\star)\}$, that is:

$$\{\lambda^l(\boldsymbol{\xi}^\star)\}_{1 \le l \le Q} = \arg \min_{\{\Lambda^l\}_{1 \le l \le Q}} \left\{ \sigma^2(\boldsymbol{\xi}^\star);\ \sum_{l=1}^{Q} \Lambda^l \mathbb{E}\{Y(\boldsymbol{\xi}^l)\} = \mathbb{E}\{Y(\boldsymbol{\xi}^\star)\} \right\}, \tag{5}$$

with:

$$\begin{aligned} \sigma^2(\boldsymbol{\xi}^\star) &= \mathrm{Var}\{Y(\boldsymbol{\xi}^\star)\} + \mathrm{Var}\{\mathscr{I}_Q Y(\boldsymbol{\xi}^\star)\} - 2\,\mathrm{Cov}\{Y(\boldsymbol{\xi}^\star), \mathscr{I}_Q Y(\boldsymbol{\xi}^\star)\} \\ &= \phi(\boldsymbol{\xi}^\star, \boldsymbol{\xi}^\star) + \sum_{l=1}^{Q} \sum_{m=1}^{Q} \Lambda^l \Lambda^m \phi(\boldsymbol{\xi}^l, \boldsymbol{\xi}^m) - 2 \sum_{l=1}^{Q} \Lambda^l \phi(\boldsymbol{\xi}^l, \boldsymbol{\xi}^\star). \end{aligned} \tag{6}$$

The covariance between Y and $\mathscr{I}_Q Y$ in Eq. (6) endows the set S_Q with an inner product and avoids the explicit knowledge of $Y(\boldsymbol{\xi})$. This technique is also known as the kernel trick and avoids explicit mappings where only the foregoing inner product between samples in S_Q is used in the evaluation of the covariance [20]. In addition, the kernel function ϕ has to be positive definite; see [18] for a comprehensive review. In the present work, the product of k one-dimensional Gaussian functions satisfying the stationary covariance process assumption has been considered. However, other kernels could be envisaged as in, e.g., [1, 18]. The covariance kernel used in the present investigation is thus given by:

$$\mathrm{Cov}\{Y(\boldsymbol{\xi}), Y(\boldsymbol{\xi}')\} = \phi(\boldsymbol{\xi}, \boldsymbol{\xi}') = \exp\left[-\frac{1}{2}\sum_{k=1}^{N}\left(\frac{\xi_k - \xi'_k}{\zeta_k}\right)^2\right], \tag{7}$$

where the kernel parameters $\boldsymbol{\zeta} = (\zeta_1, \zeta_2, \dots, \zeta_N)$ have to be computed in order to minimize the quadratic error between $\mathscr{I}_Q Y(\boldsymbol{\xi}^\star)$ and $Y(\boldsymbol{\xi}^\star)$ in Eq. (5). Here, N is the dimension of the parameters space, which is $N = 4$ in the present work.

Optimization of the Inner Parameters of the Covariance Kernel

The computation of the inner parameters of the covariance kernel is a crucial step for the accuracy of the interpolation and requires to solve a nonlinear problem in N dimensions. The choice was made to compute the kernel parameters $\boldsymbol{\zeta}$ using a cross-validation procedure [17]. The so-called leave-one-out (LOO) method consists in removing each point in the data set S_Q one by one and compute the mean-square difference E between the Kriging $\mathscr{I}_Q Y$ and its LOO counterpart $\mathscr{I}^l_{Q-1} Y$ computed from $S_Q \setminus \{Y(\boldsymbol{\xi}^l)\}$, at the location $\boldsymbol{\xi}^l$, such that:

$$E(Y)_{\mathrm{LOO}} = \frac{1}{Q}\sum_{l=1}^{Q}\left(\mathscr{I}^l_{Q-1} Y(\boldsymbol{\xi}^l) - Y(\boldsymbol{\xi}^l)\right)^2. \tag{8}$$

The computation of the Q new predictors $\mathscr{I}^l_{Q-1} Y$ can be done efficiently using Rippa's method [21]. A differential evolution algorithm has been subsequently used to solve the optimization problem for the kernel parameters $\boldsymbol{\zeta}$ associated with Eq. (8). Details about the algorithm can be found in [22].

Direct and Dual Estimate of Ordinary Kriging

Ordinary Kriging [3] is by far the most popular approach and has been considered in the present investigations. It starts with the assumption that the mean of $Y - \mathscr{I}_Q Y$ vanishes although μ is unknown. The unbiased constraint implies that:

$$\sum_{l=1}^{Q} \lambda^l(\boldsymbol{\xi}^\star)\mu - \mu = 0\,, \tag{9}$$

and thus $\sum_{l=1}^{Q} \lambda^l(\boldsymbol{\xi}^\star) = 1$. Introducing the Lagrange multiplier χ and canceling the gradient with respect to Λ^l in Eq. (6), the system to be solved, together with the condition (9), is given by:

$$\begin{aligned} 2\sum_{l=1}^{Q} \lambda^l(\boldsymbol{\xi}^\star)\phi(\boldsymbol{\xi}^l,\boldsymbol{\xi}^m) - 2\phi(\boldsymbol{\xi}^\star,\boldsymbol{\xi}^m) + 2\chi = 0\,, &\quad \forall\, m = 1,2,\dots,Q\,,\\ \sum_{l=1}^{Q} \lambda^l(\boldsymbol{\xi}^\star) = 1\,, & \end{aligned} \tag{10}$$

where the weights $\{\lambda^l(\boldsymbol{\xi}^\star)\}_{1\le l\le Q}$ are the coefficients of the linear interpolation (4) defining the Kriging surrogate. They are thus computed as the solution of the linear system:

$$\begin{bmatrix} \phi(\boldsymbol{\xi}^1,\boldsymbol{\xi}^1) & \cdots & \phi(\boldsymbol{\xi}^1,\boldsymbol{\xi}^Q) & 1\\ \vdots & \ddots & \vdots & \vdots\\ \phi(\boldsymbol{\xi}^Q,\boldsymbol{\xi}^1) & \cdots & \phi(\boldsymbol{\xi}^Q,\boldsymbol{\xi}^Q) & 1\\ 1 & \cdots & 1 & 0 \end{bmatrix} \begin{pmatrix} \lambda^1\\ \vdots\\ \lambda^Q\\ \chi \end{pmatrix} = \begin{pmatrix} \phi(\boldsymbol{\xi}^1,\boldsymbol{\xi}^\star)\\ \vdots\\ \phi(\boldsymbol{\xi}^Q,\boldsymbol{\xi}^\star)\\ 1 \end{pmatrix}. \tag{11}$$

However, this method becomes very expensive when the data set S_Q is large, and thus, numerous output values $Y(\boldsymbol{\xi}^\star)$ have to be computed. This difficulty can be overcome by observing that the linear predictor $\mathscr{I}_Q Y(\boldsymbol{\xi}^\star)$ also reads:

$$\mathscr{I}_Q Y(\boldsymbol{\xi}^\star) = \sum_{l=1}^{Q} \pi^l \phi(\boldsymbol{\xi}^l,\boldsymbol{\xi}^\star) + \pi^{Q+1}\,, \tag{12}$$

where the weights $\{\pi^l\}_{1\le l\le Q}$ are obtained as the solution of the system:

$$\begin{bmatrix} \phi(\boldsymbol{\xi}^1,\boldsymbol{\xi}^1) & \cdots & \phi(\boldsymbol{\xi}^1,\boldsymbol{\xi}^Q) & 1\\ \vdots & \ddots & \vdots & 1\\ \phi(\boldsymbol{\xi}^Q,\boldsymbol{\xi}^1) & \cdots & \phi(\boldsymbol{\xi}^Q,\boldsymbol{\xi}^Q) & 1\\ 1 & \cdots & 1 & 0 \end{bmatrix} \begin{pmatrix} \pi^1\\ \vdots\\ \pi^Q\\ \pi^{Q+1} \end{pmatrix} = \begin{pmatrix} Y(\boldsymbol{\xi}^1)\\ \vdots\\ Y(\boldsymbol{\xi}^Q)\\ 0 \end{pmatrix}. \tag{13}$$

This approach is more convenient than (4) and (11) because the weights are now independent of the unobserved shape parameters $\boldsymbol{\xi}^\star$.

Robust Optimization Based on the Surrogate Model

A tensorized grid of nine equidistributed abscissas for each of the four shape parameters of section "Shape Parameters" is used to obtain the data set S_Q; therefore, $Q = 9^4 = 6561$. The values of the lift-to-drag ratios $Y(\boldsymbol{\xi}^l)$ in S_Q as computed by the *elsA* CFD software vary in the range $[36.788, 75.198]$. The Kriging surrogate model for the present parametric problem is computed using Onera's in-house software `META_NUMF` [17]. It implements the various algorithms described in section "Kriging Surrogate Model". The inner parameters selected by the cross-validation procedure are $\boldsymbol{\zeta} \simeq (0.1020, 0.0472, 0.0957, 0.1434)$. Note that the second parameter ζ_2 (which can be interpreted as a correlation length) is rather small compared to the others. This may result in small amplitude oscillations in this direction. Also, note that the computation for the surrogate model has been carried out on eight Intel® Xeon® processors E5540 (8M cache, 2.53 GHz) for 5 days for the inner parameters optimization. The surface response obtained using the Kriging surrogate is shown in the two-parameters plan (ξ_1, ξ_3) at $\xi_2 = -0.0025c$, $\xi_4 = 0$ and $\xi_2 = 0.0025c$, $\xi_4 = 0$ in Fig. 7.

The response surface has been further characterized using a particle swarm optimization algorithm [23] to localize its extrema. The surrogate model is characterized by 28 local maxima $\{Y_m\}_{1 \le m \le 28}$, where the global maximum is $Y_{\max} \simeq 76.6724$ located at:

$$\boldsymbol{\xi}_{\max} \simeq (0.002143, -0.001873, 0.001125, 0.001162)\,,$$

such that $Y_{\max} = \mathscr{I}_Q Y(\boldsymbol{\xi}_{\max})$. The 28 local maxima extracted from the above procedure are listed below in Table 1 in 16 digits precision for illustration purposes. They proved to verify $\|\boldsymbol{\nabla}_{\boldsymbol{\xi}} Y(\boldsymbol{\xi})\| < 10^{-2}$, where the gradient was computed using a second-order centered finite difference scheme with a step $\Delta\xi_k = 0.5 \cdot 10^{-3} \times 0.0025c$ in each direction $k = 1, 2, 3$, and 4 in the design domain D. These maxima have subsequently been considered for an UQ study where the mean and the variance of each maximum have been computed conditionally upon a prior probability distribution of the shape parameters chosen as a Beta law of the first kind β_{I} for each of them.

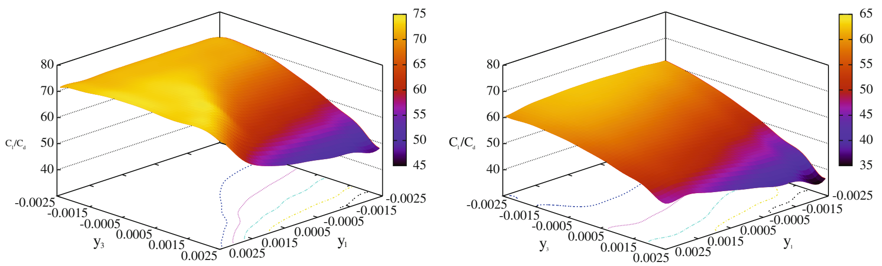

Fig. 7 Surface response in the (ξ_1, ξ_3) plan for $\xi_2 = -0.0025$ (left) and $\xi_2 = 0.0025$ (right) with $\xi_4 = 0$ in both cases

Table 1 List of local maxima $\{Y_m\}_{1\leq m\leq 28}$ identified on $\mathscr{I}_Q Y(\boldsymbol{\xi})$

m	ξ_1	ξ_2	ξ_3	ξ_4	$Y(\boldsymbol{\xi})$
1	0.0009247281603877587	-0.002477328481440417	-0.0020522618807618394	-0.001006637007272001	72.96062013330138
2	-0.0019877340165556532	-0.0012501286098216477	-0.0022230288250307834	1.637485303221777e-05	72.9053124038243
3	-0.0021080331586727672	-0.0024760806714095025	-0.0022790922661521593	-0.0009996692634806229	72.80830395616871
4	0.0009227995269777299	-0.0012491170657324849	-0.002001537347927156	0.0003111090101579929	73.11460103259566
5	0.0014184152803302856	-0.002481992250078206	0.0009674096676959162	-0.0013436457334901153	75.13385342547113
6	0.0021005751944998536	-0.0024838327139435935	0.0016378831262294602	-0.0007521259615468634	76.19703358554321
7	0.002260824716209251	-0.001875360609138144	-0.000147906043850845	0.0011438578569142157	75.33061972327575
8	0.001495913190949574	-0.0018758598395562961	0.0015100035322869972	-0.0014295665683347527	74.01965083298452
9	0.0007217459064240732	5.673767819445487e-07	-0.002037392586569452	0.0007255817623195416	73.37227470479218
10	-0.0020511434400430797	-0.0012442591298048122	-0.0012727785963404998	0.000999942602579444	72.48039437600411
11	-0.00046863745004457153	-7.173866486613438e-07	-0.002172688079996433	0.0003997664586228222	73.63518562977276
12	0.00011892753917269912	-0.002479381700009219	-0.0007452975881961537	-0.002277292293674497	74.32636561284727
13	0.0021112462457524967	-0.0012465346002510994	0.001580251301382974	0.0016964375010017608	76.1863078631464
14	0.002144283961154753	-0.0018726431452980397	0.00112667245068374	0.0011645115201617874	76.67235940824781
15	0.0009176228156363136	-0.0012482712347895557	0.0004398328525607037	0.0006259803524756003	75.76076285356561
16	0.002252700336863187	4.600109478019969e-07	-0.0007018136569205303	0.0022622430964318965	75.04867663515955
17	-4.1945374307096866e-05	0.0006262230074350205	-0.0021470974826768028	0.001330451966658052	73.65690479624587
18	0.0008007997438743406	-0.002481211158065615	0.00032503601521170766	-0.002393635032754219	74.99121577439612
19	0.00027683857843118897	-0.0012451399796126398	-8.168292353796704e-05	0.00041319595122059957	75.42284453664654
20	0.0010060672108371622	-0.0006244922023885663	-0.0007618898135975365	0.0011409171442569492	74.47172633897203
21	0.002129228704884845	-0.0012506867133831355	0.000426988917664922	0.0017446918994712614	75.34062327397848
22	0.0010563407826690357	0.0006259797249405174	-0.0010642802734270236	0.0021843136713695587	74.19290076614386
23	-0.0003057993918150065	-0.001247433430195173	-0.0007151826244908299	-4.83642984228606e-05	74.68696951011597
24	0.0009907930610100206	0.001875515814836005	-0.0022052419551779243	0.0022553625771607956	74.0466960120915
25	0.0013825430485137993	-0.0018710126427371548	0.0005284653042229091	0.0004806569969429608	75.84927930947487
26	-0.0010037161479930269	-0.0012503314535020984	-0.002172701898798983	-0.0008923292318644253	73.695550780483
27	0.002240997891465996	0.0018755852596172282	-0.0021457313740432917	0.0023055559834000336	74.34508504051362
28	0.002293961593751379	-0.0006239914756562181	-0.00018841428563772227	0.0021561109756890805	74.97687454582714

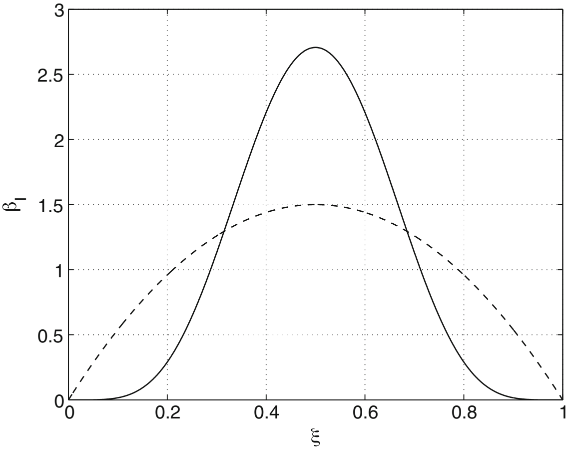

Fig. 8 β_I probability distribution functions for $\alpha = \beta = 6$ (full line) and $\alpha = \beta = 2$ (dashed line) used for modeling the uncertainty on the design of the shape parameters

This distribution function appears as a suitable choice when considering manufacturing defects since it has a compact support. In addition, it is the law arising from Jaynes' maximum entropy principle [24] if the averages $\mathbb{E}\{\log \xi\}$ and $\mathbb{E}\{\log(1-\xi)\}$ are given. The convention for the definition (on the normalized parameter domain $\hat{D}$) of the Beta distribution β_I used in the present study is:

$$\beta_\mathrm{I}(\xi;\alpha,\beta) = \mathbb{1}_{[0,1]}(\xi)\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\xi^{\alpha-1}(1-\xi)^{\beta-1}\,, \tag{14}$$

where $z \mapsto \Gamma(z)$ is the usual Gamma function defined by $\Gamma(z) = \int_0^{+\infty} t^{z-1}\mathrm{e}^{-t}dt$. The probability density functions are taken symmetric with $\alpha = \beta = 6$ for the bump located near the leading edge at 5% of the chord c (parameter ξ_1) and $\alpha = \beta = 2$ for the three other bumps (parameters ξ_2, ξ_3 and ξ_4); see Fig. 8. The amplitude of the uncertainty was taken in the range $\pm 5 \cdot 10^{-4}c$ which is consistent with observations from the industry [25]. The present mean and variance are computed by a Monte Carlo procedure using $n_s = 10^4$ samples. Given the convergence of the Monte Carlo method as $\mathrm{O}(1/\sqrt{n_s})$, the statistics have been rounded to two significant digits in Table 2.

The solution of the UQ is shown in Table 2 for the maxima identified in Table 1. The mean of the maxima appears to be lower than the maximum found during the optimization process with variances in the range $[0.1, 1.2]$. We define the robust maximum Y_R such that:

$$Y_\mathrm{R}(\boldsymbol{\xi}) = \mathbb{E}\{Y(\boldsymbol{\xi})\} - 3\sqrt{\mathrm{Var}\{Y(\boldsymbol{\xi})\}} \tag{15}$$

has the largest value. Based on that definition, the maximum identified during the optimization procedure (labeled $m = 14$ in Table 1 and highlighted in gray) is not the robust one. Indeed, for the global deterministic maximum, $Y_{\max} \simeq 76.67$ that decreases to $\mathbb{E}\{Y_{14}\} \simeq 75.5$ with $\sqrt{\mathrm{Var}\{Y_{14}\}} \simeq 0.74$ which gives $Y_{\mathrm{R},14}(\boldsymbol{\xi}) \simeq 73.28$.

Table 2 List of mean maxima of $\mathscr{I}_Q Y(\boldsymbol{\xi})$ and their variance for the uncertainty propagation procedure described in section "Robust Optimization Based on the Surrogate Model"

m	ξ_1	ξ_2	ξ_3	ξ_4	$\mathbb{E}\{Y(\boldsymbol{\xi})\}$	$\sqrt{\mathrm{Var}(Y(\boldsymbol{\xi}))}$
1	0.00092472816038	-0.00247732848144	-0.00205226188076	-0.00100663700727	72.22	0.51
2	-0.0019877340165	-0.00125012860982	-0.00222302882503	1.63748530322e-05	72.24	0.27
3	-0.0021080331586	-0.00247608067141	-0.00227909226615	-0.00099966926348	71.99	0.49
4	0.00092279952697	-0.00124911706573	-0.00200153734793	0.000311109010158	72.54	0.26
5	0.00141841528033	-0.00248199225008	0.000967409667696	-0.00134364573349	74.01	0.87
6	0.00210057519450	-0.00248383271394	0.001637883126230	-0.00075212596154	74.81	1.11
7	0.00226082471621	-0.00187536060914	-0.00014790604385	0.001143857856910	74.59	0.39
8	0.00149591319095	-0.00187585983956	0.001510003532290	-0.00142956656833	73.10	0.44
9	0.00072174590642	5.67376781945e-07	-0.00203739258657	0.000725581762320	72.78	0.27
10	-0.0020511434400	-0.00124425912980	-0.00127277859634	0.000999942602579	71.66	0.35
11	-0.0004686374500	-7.1738664866e-07	-0.00217268808000	0.000399766458623	72.98	0.30
12	0.00011892753917	-0.00247938170001	-0.00074529758819	-0.00227729229367	73.38	0.70
13	0.00211124624575	-0.00124653460025	0.001580251301380	0.001696437501000	74.87	0.83
14	0.00214428396115	-0.00187264314530	0.001126672450680	0.001164511520160	75.50	0.74
15	0.00091762281563	-0.00124827123479	0.000439832852561	0.000625980352476	74.77	0.57
16	0.00225270033686	4.60010947802e-07	-0.00070181365692	0.002262243096430	74.29	0.39
17	-4.194537430e-05	0.000626223007435	-0.00214709748268	0.001330451966660	73.00	0.30
18	0.00080079974387	-0.00248121115807	0.000325036015212	-0.00239363503275	73.85	0.86
19	0.00027683857843	-0.00124513997961	-8.1682923538e-05	0.000413195951221	74.47	0.54
20	0.00100606721084	-0.00062449220238	-0.00076188981359	0.001140917144260	73.81	0.35
21	0.00212922870488	-0.00125068671338	0.000426988917665	0.001744691899470	74.50	0.46
22	0.00105634078267	0.000625979724941	-0.00106428027343	0.002184313671370	73.52	0.34
23	-0.0003057993918	-0.00124743343020	-0.00071518262449	-4.8364298229e-05	73.91	0.42
24	0.00099079306101	0.001875515814840	-0.00220524195518	0.002255362577160	73.33	0.31
25	0.00138254304851	-0.00187101264274	0.000528465304223	0.000480656996943	74.84	0.57
26	-0.0010037161479	-0.00125033145350	-0.00217270189880	-0.00089232923186	73.04	0.29
27	0.00224099789147	0.001875585259620	-0.00214573137404	0.002305555983402	73.58	0.32
28	0.00229396159375	-0.00062399147565	-0.00018841428563	0.002156110975691	74.23	0.37

On the other hand, the local robust maximum is the point labeled $m = 7$ in Table 2 and highlighted in light gray, for which $Y_7 \simeq 75.33$, $\mathbb{E}\{Y_7\} \simeq 74.59$ with a variance of $\sqrt{\mathrm{Var}\{Y_7\}} \approx 0.39$ and $Y_{\mathrm{R},7}(\boldsymbol{\xi}) \simeq 73.42$. This surrogate model appears therefore as an interesting test case for robust optimization as the global maximum, when subjected to uncertainty, happens to be less robust than a local maximum which maximizes the mean of the quantity of interest and minimizes the standard deviation associated with a stochastic process.

Conclusions

In this chapter, we have addressed an uncertainty quantification and shape optimization problem for a RAE 2822 airfoil at transonic speeds using a Kriging-based surrogate model of the lift-to-drag ratio. More specifically, the shape of the baseline profile has been altered by four localized bumps of variable amplitudes on its suction side. Their locations have been selected so that they modify the position of the shock which primarily drives the aerodynamic performance of the profile, such as the lift and drag forces. The transonic flow about the airfoil has been simulated using the Onera software *elsA* for each instance of the bump amplitudes in a tensorized grid with nine equidistant sampling values for each amplitude parameter, resulting in a dense data set of 9^4 samples. Each simulation has been performed for carefully chosen numerical parameters optimized on the basis of the baseline configuration compared with the available experimental results. The high-quality, dense Kriging surrogate model interpolating the data set uses Gaussian covariance kernels and exhibits multiple local maxima. They have been identified by a particle swarm algorithm, although its results might not be exhaustive and more maxima might be found using an alternative algorithm.

An uncertainty quantification analysis has subsequently been carried out about each local maxima, assigning a variation range and a prior probability density function to the shape parameters to mimic the effect of design tolerances and aging. This analysis has revealed that the global maximum, for example, is not necessarily the most robust one in terms of its standard deviation or of a robust design criterion such as the mean minus three times the standard deviation. It is thus believed that the data set and results obtained in the present work constitutes an interesting test case for assessing uncertainty quantification and/or robust optimization strategies in future investigations.

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